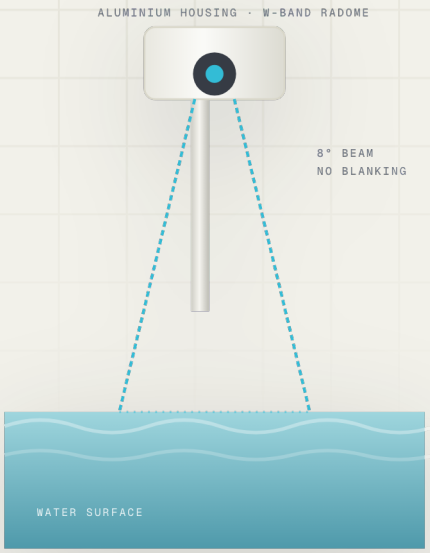
VEPOLINK
MONITORING PURITY

PROCESS INSTRUMENTATION × WATER LEVEL

NEXT-GENERATION RADAR, SELF-MONITORING BUILT IN

RadarSense *Smart Level Sensor*

An 80 GHz FMCW radar sensor for reliable, non-contact water level measurement — with no blanking distance, built-in tilt and humidity monitoring, and a low total cost of ownership.



0–30 m
MEASUREMENT RANGE

±2 mm
LEVEL ACCURACY

80 GHz
FMCW RADAR FREQUENCY

80 GHZ FMCW NO BLANKING DISTANCE | **SMART SENSORS** TILT & HUMIDITY |
VEPOLINK FIELD-PROVEN BUILD

0m
BLANKING DISTANCE, MOUNT CLOSE TO WATER

±2mm
ACCURACY ACROSS THE FULL 0-30M RANGE

4mA
ACTIVE CURRENT DRAW

IP67
FLOOD-PROOF, 1 WEEK / 1M SUBMERSION

01 OVERVIEW

Radar that *watches itself too.*

An 80 GHz FMCW radar sensor engineered for reliable, non-contact water level measurement. The higher frequency gives a narrower 8° beam and no blanking distance at all, so the sensor can be mounted close to the water and still read accurately.

Smart sensing adds built-in inclination monitoring on two axes and an internal humidity sensor, so drift or condensation inside the housing shows up remotely — long before it affects a reading — cutting the number of trips to the field.

REPORTED CHANNELS

- WATER LEVEL / DEPTH TO WATER
- INTERNAL RELATIVE HUMIDITY
- SENSOR ORIENTATION (TILT)
- SDI-12 OUTPUT
- RS-485, MODBUS RTU
- HYDROLOGICAL DISCHARGE (Q)
- INTERNAL DATA LOGGER

· PRODUCT FEATURES

80 GHz FMCW Radar
A higher radar frequency delivers unmatched accuracy and a narrow 8° beam, with no blanking distance to work around.

Built-In Health Monitoring
An X/Y inclination sensor and an internal humidity sensor flag mounting drift or condensation remotely, before they affect data.

On-Board Value Processing
Mean, minimum, maximum, median, standard deviation and hydrological discharge (Q) are all computed on the sensor itself.

Flood-Proof Housing
IP67-rated, with the lower housing tolerating full submersion for up to one week at a 1 m water column.

02 MEASUREMENT PRINCIPLE

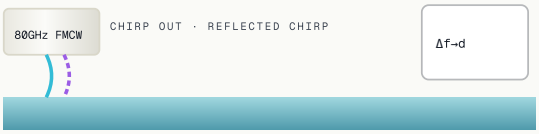
FMCW radar, *measured in frequency.*

A frequency-modulated continuous wave signal in the 77–81 GHz band is transmitted toward the water. Distance is derived by comparing the frequency of the reflected signal to the signal being transmitted at that same instant — an indirect transit-time measurement with no dead zone close to the sensor.

FMCW

Frequency-Based Ranging

77–81 GHZ · W-BAND · CONTINUOUS WAVE



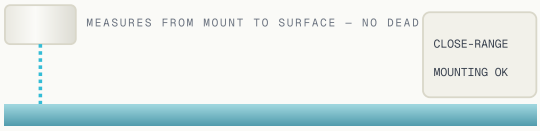
Comparing transmitted and reflected frequencies gives a highly precise distance reading, independent of signal amplitude.

`Δf → distance`

MOUNTING

No Blanking Distance

CLOSE MOUNTING · 8° BEAM ANGLE



Unlike a wider dead zone near the sensor, there is no minimum distance below which the sensor cannot read — close mounting is fully usable.

`Mount close, read anyway`

SMART SENSING

Self-Monitoring Built In

X/Y INCLINATION · INTERNAL HUMIDITY



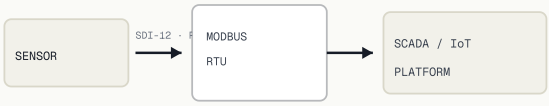
Internal inclination and humidity sensors report device health alongside the level reading, reducing unplanned field visits.

`Device health, remotely`

OUTPUT

Transmission & Reporting

SDI-12 · RS-485 (MODBUS RTU)



Standard SDI-12 and Modbus RTU interfaces connect directly into existing telemetry and SCADA platforms.

`Standard protocols, no extras`

03 SMART DIAGNOSTICS

Knowing the sensor *is still reading true.*

A radar sensor that silently drifts out of plumb, or quietly accumulates condensation, can keep reporting numbers without flagging that anything is wrong. Built-in diagnostics close that gap.

SENSOR 01

X/Y inclination

Two-axis tilt sensing reports mounting angle, flagging bracket shift or pole movement before it skews the reading.

SENSOR 02

Internal humidity

An internal relative humidity sensor reports condensation risk inside the enclosure ahead of any electronic impact.

ACCESS

SDI-12 & Modbus registers

Both diagnostic values are retrievable over the same interface as the level reading — no separate diagnostic channel.

ON-BOARD VALUE PROCESSING

The sensor computes mean, minimum, maximum, median and standard deviation over a time period directly on-board, along with hydrological discharge (Q) where a rating relationship is configured — reducing what downstream systems need to calculate.

FEWER TRIPS TO THE FIELD

Because tilt and humidity are visible remotely alongside the level reading, a technician can confirm the sensor is sound before deciding whether a visit is actually needed — particularly valuable at hard-to-reach or flood-prone sites.

SAMPLING RATE

The standard sensor scans the water surface within one second per reading, giving a steady 2 Hz sampling option suited to routine stage monitoring and flood warning networks.

04 SPECIFICATIONS

Technical *specifications.*

MEASUREMENT

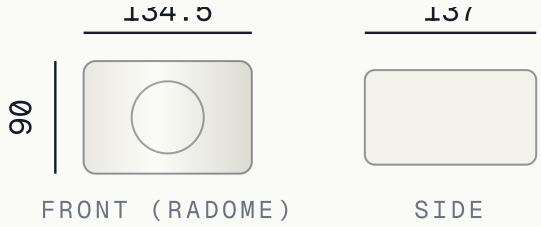
Parameter	Water level / depth to water
Working Principle	FMCW radar, 80 GHz (W-band)
Measuring Range	0 – 30 m, distance to water
Level Accuracy	± 2 mm
Resolution	0.001 m · 0.1 cm · 1 mm
Beam Angle	8°, no blanking distance
Sampling Option	2 Hz standard
Additional Parameters	Internal humidity · sensor orientation
Value Processing	Mean · min · max · median · std. dev. · discharge (Q)

SYSTEM · BUILD · POWER

Interface	SDI-12 & RS-485 (Modbus RTU)
Internal Data Logger	Yes
Power Supply	5.5 – 28.8 V DC, typ. 12/24 V DC
Power Consumption, Active	4 mA
Power Consumption, Idle	250 µA
Housing Material	Aluminium (AlMgSi1); ASA / PTFE radome
Mounting Material	Stainless steel 1.4301 (V2A)
Cable	PUR jacket, M9 connector
Protection Class	IP 67 – floods without damage, 1 week / 1 m water column
Dimensions (H×W×D)	90 × 134.5 × 137 mm

SENSOR HEAD DIMENSIONS

INDICATIVE FRONT & SIDE · MM



FRONT (RADOME) SIDE

05 INSTALLATION & SITING

Mount close, *read accurately anyway.*

> Because there is no blanking distance, the sensor can be mounted lower and closer to the water than older radar designs allow — useful where mounting height is limited, such as on a short bridge rail or a low pier. It carries an **IP67 rating** and can be mounted outdoors in fully unprotected locations.

01 · MOUNTING POINT

Bridge, pier or arm

Mount above the expected maximum water level, aimed straight down at the measurement point.

02 · OBSTRUCTION CHECK

Low water inspection

Check for submerged obstructions such as rocks or piers at the lowest anticipated water level, before they are covered.

03 · SURFACE SMOOTHNESS

Minimise wave action

Choose a site where the water surface stays as smooth as possible within the 8° beam footprint.

04 · ORIENTATION CHECK

Use the inclination sensor

Read the built-in X/Y tilt values during commissioning to confirm the mount is level and stays that way over time.

05 · INTERFACE WIRING

SDI-12 or Modbus RTU

Wire into the site's datalogger or telemetry unit over SDI-12 or RS-485, matching whichever protocol the platform already uses.

06 · AVERAGING SETUP

Configure the measurement interval

Set the measurement interval and averaging time appropriate to the site's flow dynamics during commissioning.

REMOTE HEALTH CHECKS

Once commissioned, tilt and humidity values can be polled remotely at any time, giving an early warning if the mount shifts or the enclosure seal is compromised — well before the level reading itself is affected.

06 APPLICATIONS

Where it *works.*

Networks that need both high accuracy and confidence that every sensor in the field is still reading true, without a dedicated visit to check.

Flood Monitoring & Warning	Tide & Coastal Monitoring	Storm Water Systems
Rain Retention Basins	Reservoirs & Dams	Drought Monitoring

STREAM GAUGING NETWORKS

With discharge (Q) computed on-board against a configured rating curve, the sensor fits directly into existing stream gauging and hydrological monitoring networks without extra processing downstream.

COMPACT, LOW-CLEARANCE SITES

No blanking distance makes this the practical choice where mounting height above the water is limited, such as low bridges, culvert crowns or shallow retention structures.

07 FAQ

Frequently asked *questions.*

Q What is the working principle?

Frequency-modulated continuous wave (FMCW) radar at 80 GHz, comparing transmitted and reflected signal frequency to derive distance.

Q What does "no blanking distance" mean?

There is no minimum dead-zone distance near the sensor below which it cannot read — it can be mounted closer to the water than older radar designs allow.

Q How accurate is it?

±2 mm across the full 0–30 m measuring range.

Q What do the inclination and humidity sensors do?

They report mounting tilt on two axes and internal condensation risk remotely, flagging issues before they affect the level reading.

Q Can it survive flooding?

Yes — the housing is IP67 rated, tolerating full submersion for up to one week at a 1 m water column.

Q What outputs are available?

SDI-12 and RS-485 carrying Modbus RTU, both over the same interface as the level reading.

Q What does it compute on-board?

Mean, minimum, maximum, median and standard deviation over a configurable time period, plus hydrological discharge (Q) where a rating curve is set.

Q How often does it sample?

The standard sensor scans the water surface within one second, giving a steady 2 Hz sampling option.

Q What is the housing made of?

Aluminium (AlMgSi1) with an ASA/PTFE radome and stainless steel (1.4301) mounting hardware.

Q What maintenance does it need?

Minimal — remote tilt and humidity checks largely replace routine physical inspection visits.

This brochure is general product information. Configuration, dimensions and performance figures are indicative and confirmed at order. For installation guidelines and full technical detail, refer to the user manual supplied with the product. Specifications may change without notice as the product is improved.

VEPOLINK
MONITORING PURITY

RADAR THAT TELLS YOU WHEN SOMETHING'S OFF

Know your level. *Know your sensor too.*

Tell us the site's range, mounting height and telemetry platform, and we'll confirm the fit and configure the output. Pick how you'd like to start.

01 **Talk to our engineers**
Speak directly to the team that sizes, specifies and commissions smart radar level sensors. No call centre.
+91 99538 86178 →

02 **Request a site sizing** RECOMMENDED
Send your expected range, mounting height and telemetry platform — we'll confirm the fit and output.
sales@vepolink.com →

03 **Book a demonstration**
See a running sensor and its diagnostics with our engineering team.
Request a demo →